QUANTUM MACHINE LEARNING FOR FLOOD SUSCEPTIBILITY MAPPING

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ABSTRACT

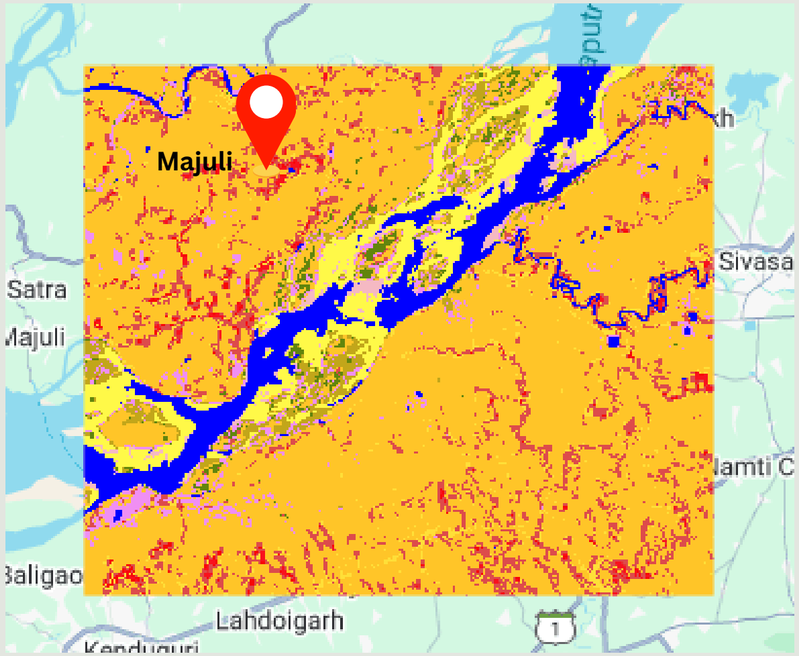
Recurrent flooding in the Brahmaputra Basin of Assam causes significant socio-economic and agricultural losses, highlighting the need for accurate flood susceptibility mapping. This study explores the applicability of Quantum Machine Learning (QML) for flood susceptibility prediction using satellite-derived geospatial features. A classical Random Forest model is compared with quantum models, including QSVM, VQC, and QNN. Model performance is evaluated using accuracy, precision, recall, and ROC–AUC, highlighting current capabilities and limitations of QML on real geospatial data.

INTRODUCTION

Flooding is a recurring natural hazard in the Brahmaputra Basin of Assam, causing extensive damage to lives, agriculture, and infrastructure. Accurate flood susceptibility mapping is essential for effective risk management and planning. Machine Learning techniques have been widely used to model complex, nonlinear relationships in geospatial flood data, offering improved predictive capabilities. Recently, Quantum Machine Learning has emerged as a novel paradigm, potentially enhancing learning through quantum feature representations. This study explores and compares classical and quantum models for flood susceptibility mapping.

PROBLEM

The Brahmaputra Basin in Assam experiences frequent and severe flooding, yet reliable flood susceptibility maps remain limited due to complex geospatial interactions. Conventional methods and classical models face challenges in scalability and feature complexity, motivating the evaluation of Quantum Machine Learning for improved flood susceptibility prediction.



Majuli, Assam, India - LULC

OBJECTIVE

- To explore the applicability of Quantum Machine Learning (QML) for flood susceptibility prediction and compare its performance against a classical Machine Learning baseline.
- Specifically:
- Build a flood susceptibility model using geospatial features derived from satellite remote sensing.
- Evaluate and compare Random Forest (classical) vs QSVM, VQC, QNN (Quantum ML).
- Investigate whether QML shows performance advantages or limitations on real geospatial data

METHODOLOGY

DATASET CREATION

Dataset Creation using Google Earth Engine

Features sampled (2981 points, 100m resolution):

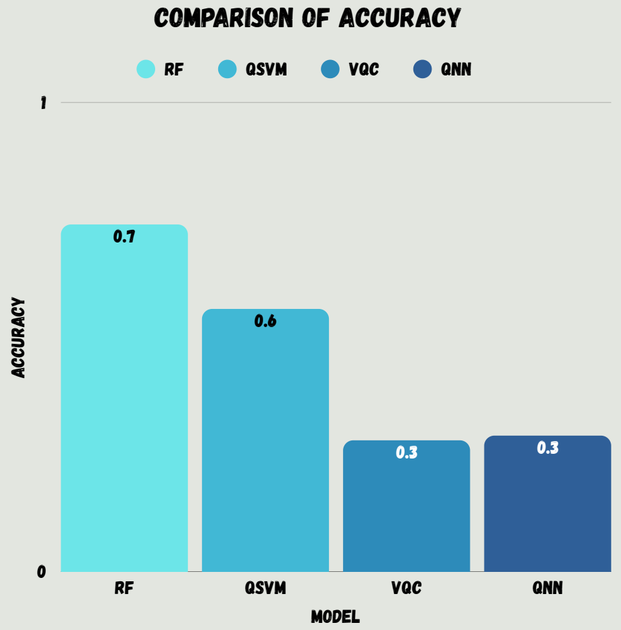
- Elevation (DEM - SRTM)
- Slope, Aspect
- Distance to water
- LULC (Copernicus)
- Flooded / Not Flooded label (Global Flood DB 2000–2018)

Dataset Preview

	system:index	aspect	dist_to_water	elev	flooded	lulc	slope
0	0	312	33.136083	99	0	40	3
1	1	156	7.071068	97	1	60	5
2	2	270	67.364679	89	0	40	1

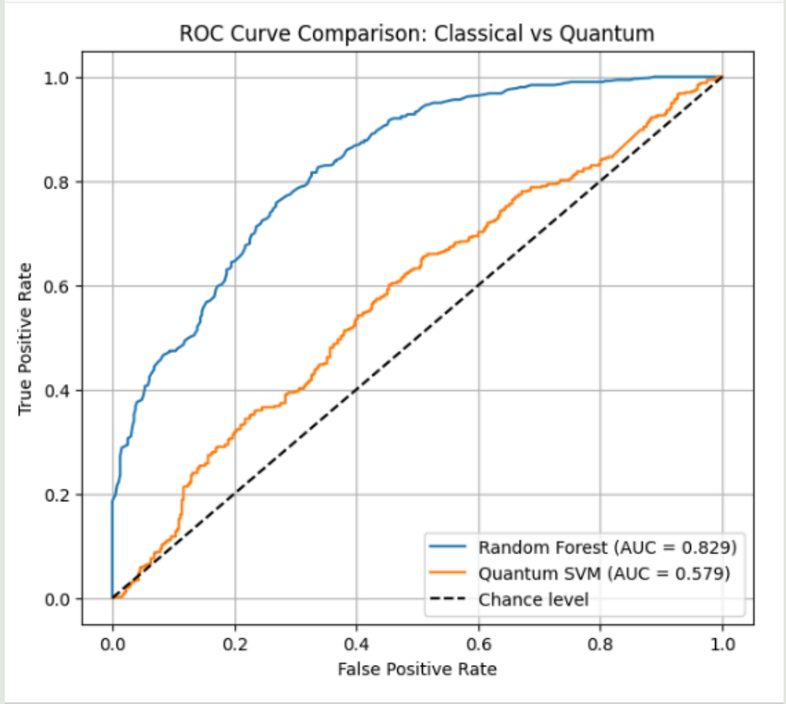
RESULT

- RF performed best (0.74) - strong for tabular nonlinear features
- QSVM moderate (0.56) - learning above chance level
- VQC, QNN low (<0.30) - optimization failure & barren plateaus
- Classical > Quantum for now, but QML promising for future scaling



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CONCLUSION

Quantum ML does not outperform classical methods yet, but provides valuable insights into limitations & potential of hybrid computational approaches for flood prediction.

FUTURE WORK:

- Run on IBM Quantum backend
- Add hydrology/soil/rainfall data
- Improve VQC architecture & optimizers

REFERENCE

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Thank You

For your attention